



AKTU 2025-26

B.Tech / MCA



DSTL (CS & CS Allied)

Most Imp Topics



Discrete Structures and Theory of Logic (DSTL) OR Discrete Mathematics

UNIT - 1 : (i) Set Theory (ii) Relations (iii) POSET (iv) Lattices

✓ **UNIT - 2 :** (i) Functions (ii) Boolean Algebra

✓ **UNIT - 3 :** Theory of Logics : (i) Proposition Logic (ii) Predicate Logic

UNIT - 4 : Algebraic Structures

✓ **UNIT - 5 :** (i) Graphs (ii) Combinatorics

UNIT-1 : Set Theory & Relations , POSET & Lattices

➤ Laws on Set theory

➤ Properties of Relation

➤ POSET (Hasse Diagram)

➤ LATTICES

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UNIT- 2 : Functions & Boolean Algebra

- Types of function
 - Composite function
 - Theorems of Boolean algebra
 - CN form and DN form
 - K-MAP V.V.Imp
- 7
- 7
- 7

UNIT – 3 : Theory of Logics

(i) Proposition Logic

- Truth tables, Tautology, Satisfiability and Contradiction
- Theory of Inference

(ii) Predicate Logic

- Quantifiers
- Inference theory of predicate logic.

DSTL : Important Topics

UNIT- 4 : Algebraic Structures

➤ Question based on Groups, Subgroups, Cyclic Groups,

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➤ Normal Subgroups, Cosets and Lagrange's theorem

➤ Group Homomorphism's

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➤ Elementary properties of Rings and Fields

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Unit - 5 : Graphs and Combinatorics

(i) Graphs

- Basic terminology
- Types of Graphs and Related theorems 7
- Isomorphism and Homeomorphism of graphs
- Graph coloring

(ii) Combinatorics

- Counting Techniques
- Pigeonhole Principle 7



Gateway Classes

Empowering Learners, Transforming Futures



Thank You



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